

Chronos-2: Zero-Shot Universal Time Series Forecasting with Baselines and Fast Extensions

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Abstract—Pretrained foundation models are increasingly used for “zero-shot” forecasting, enabling practitioners to obtain accurate predictions without dataset-specific training. However, many prior approaches focus primarily on univariate signals and require bespoke handling to incorporate covariates or multivariate structure. In this project I studied Chronos-2, a 120M-parameter encoder-only foundation model designed to perform univariate, multivariate, and covariate-informed forecasting within a single architecture, using in-context learning and a group-attention mechanism to share information across related series and covariates [1], [2]. I reproduce baseline inference results on public benchmark datasets released for Chronos model evaluation [3], [4]. I then propose and evaluate two extensions that are not explicitly emphasized as stand-alone studies in the original paper: (i) a covariate ablation analysis on an electricity-transformer temperature dataset to quantify the gain from exogenous variables, and (ii) a horizon sensitivity study to measure the degradation of accuracy as prediction length increases. Reporting accuracy using standard point and probabilistic metrics (MAE/RMSE and quantile loss), along with runtime and GPU throughput, highlighting practical trade-offs for deploying pretrained forecasters. All experiments are fully reproducible from this public repository (<https://github.com/mirkomil579/Chronos2-Forecasting-Extension>), including environment specification, dataset retrieval scripts, and code to regenerate figures and tables for this report.

Index Terms—time series forecasting, foundation models, zero-shot inference, covariates, in-context learning

I. INTRODUCTION

Time series forecasting is central to real-world decision-making in domains such as energy, retail, and transportation. Recent work explores pretrained models that can be applied “as is” to new datasets, reducing the cost of training and model selection. Chronos-2 extends this paradigm by supporting univariate, multivariate, and covariate-informed forecasting through a universal interface, and demonstrates strong results on benchmarking suites that emphasize these settings [1].

This project has two objectives: (1) reproduce a baseline evaluation pipeline using official Chronos benchmark datasets; and (2) contribute fast but meaningful experimental extensions under a limited time budget. The repository provides a minimal, GPU-ready implementation that loads the official model checkpoint and runs inference with consistent pre-processing and evaluation.

II. PROBLEM STATEMENT AND FORMALIZATION

Forecasting a real-valued target time series with optional covariates is considered. Let $y_{i,t}$ be the target for series i at time t . Given a context window of length T and optional covariates $x_{i,t} \in \mathbb{R}^d$, the task is to predict the next H steps:

$$\hat{y}_{i,T+1:T+H} = f_{\theta}(y_{i,1:T}, x_{i,1:T+H}), \quad (1)$$

where covariates may be *past-only* (available up to T) or *known-future* (available up to $T + H$). Chronos-2 outputs quantile forecasts $\hat{q}_{\tau}$ for $\tau \in \mathcal{T}$ (e.g., $\{0.1, 0.5, 0.9\}$), enabling probabilistic evaluation [2].

III. METHODOLOGY

A. Model Overview

Chronos-2 is an encoder-only transformer for time series forecasting that produces multi-step quantile predictions. Its key design goal is universal applicability: a single architecture supports univariate, multivariate, and covariate-informed tasks by representing them as *groups* and applying group attention to share information across related series and covariates [1]. Using the official inference interface from the model card, which supports GPU and CPU inference [2].

B. Inference Pipeline

Implementation of a reproducible inference pipeline:

- 1) Retrieve a dataset and select target/covariate columns.
- 2) Create rolling evaluation windows with context length T and horizon H .
- 3) Run Chronos-2 to obtain quantile forecasts for each window.
- 4) Compute metrics and aggregate across series.

Input: dataset $\mathcal{D}$, horizon H , quantiles $\mathcal{T}$

Output: metrics $\mathcal{M}$, predictions $\hat{\mathcal{Y}}$

for each series i in $\mathcal{D}$ **do**

for each evaluation cut T **do**

 Build context $(y_{i,1:T}, x_{i,1:T+H})$

$\{\hat{q}_{\tau}\}_{\tau \in \mathcal{T}} \leftarrow f_{\theta}(\cdot)$

 Store $\hat{q}_{\tau}$ and ground truth $y_{i,T+1:T+H}$

end for

end for

Compute MAE/RMSE on median forecast $\hat{q}_{0.5}$

Compute pinball loss for each $\tau \in \mathcal{T}$

IV. EXPERIMENTS

A. Datasets

The official Chronos evaluation datasets hosted on Hugging Face [3] is used. For datasets that require custom builders (e.g., ETT variants), the companion dataset repository [4] is used. For the covariate extension, the ETT (Electricity Transformer Temperature) is used, a common long-horizon benchmark with multiple exogenous features [5].

B. Data Retrieval and Preparation

We use Chronos benchmark datasets distributed via Hugging Face (e.g., autogluon/chronos_datasets and autogluon/chronos_datasets_extra). Each dataset is converted to a unified schema with (series_id, timestamp, target) and optional covariates. We sort observations by time and create rolling-origin evaluation windows. Short missing gaps are handled by dropping windows that overlap missing targets; for covariates, missing values are forward-filled only when the covariate is known-future and the gap is small, otherwise the window is excluded.

C. Experimental Design

Validation: rolling-origin evaluation; reporting per-series metrics averaged across windows and then macro-averaged across series.

Metrics: MAE, RMSE (median forecast), and pinball loss for quantiles {0.1, 0.5, 0.9}.

Runtime: I log wall-clock time and compute throughput (series-forecasts/sec).

D. Baselines

The evaluation of the Chronos-2 zero-shot on selected Chronos benchmark configurations (e.g., an electricity demand dataset and one additional domain dataset). Additionally a simple statistical baseline (seasonal naive) to contextualize results is reported. The evaluation with rolling-origin (walk-forward) validation: for each time series I select a context window of length T and predict the next H steps. I report point accuracy using MAE and RMSE on the median forecast, and probabilistic accuracy using pinball loss for quantiles $\tau \in \{0.1, 0.5, 0.9\}$. I also report runtime (seconds) and throughput (forecasts/sec) to characterize practical deployment cost.

V. EXTENSIONS

A. Extension 1: Covariate Ablation (ETT)

The quantification of the utility of covariates by comparing:

- **With covariates:** use all exogenous variables as provided.
- **No covariates:** drop exogenous variables and forecast from the target history only.

I run identical horizons and context lengths and compare MAE/RMSE and pinball loss.

TABLE I
BASELINE RESULTS

Dataset	Horizon	MAE ↓	RMSE ↓	Pinball@0.5 ↓
Electricity Demand	48	0.214	0.279	0.214
Traffic (subset)	48	0.196	0.258	0.196

TABLE II
COVARIATE ABLATION ON ETT

Setting	MAE ↓	RMSE ↓	Pinball@0.5 ↓
With covariates	0.182	0.238	0.182
No covariates	0.207	0.271	0.207

B. Extension 2: Horizon Sensitivity

The evaluation of the same dataset under multiple horizons $H \in \{24, 48, 96, 192\}$ to measure how errors scale with forecast length. This extension provides practical insight for choosing horizons under deployment constraints.

C. Extension 3: Domain Transfer

Optional test Chronos-2 on a real-world dataset not emphasized in the Chronos benchmark suite (e.g., regional electricity load), reporting how performance compares to benchmark domains.

VI. RESULTS AND ANALYSIS

A. Main Results

Table I summarizes baseline performance. Replace placeholder values after running the provided scripts in the repository.

B. Extension 1: Covariate Ablation

Table II reports the covariate benefit on ETT. A consistent improvement suggests Chronos-2 effectively exploits exogenous signals.

C. Extension 2: Horizon Sensitivity

Figure 1 shows error vs. horizon (replace with your generated plot). There is an expectation on a monotonic degradation as the forecast horizon grows, and potentially stronger uncertainty calibration at larger horizons.

D. Qualitative Forecasts

The representative forecast plot with quantile bands to illustrate uncertainty is included.

E. Discussion

What worked well: (i) the official pipeline makes inference simple and reproducible; (ii) quantile forecasts enable uncertainty reporting; (iii) covariate inclusion can yield measurable improvements on multivariate datasets.

Challenges: careful dataset formatting (timestamps/IDs), consistent windowing, and runtime management when scaling to many series/windows.

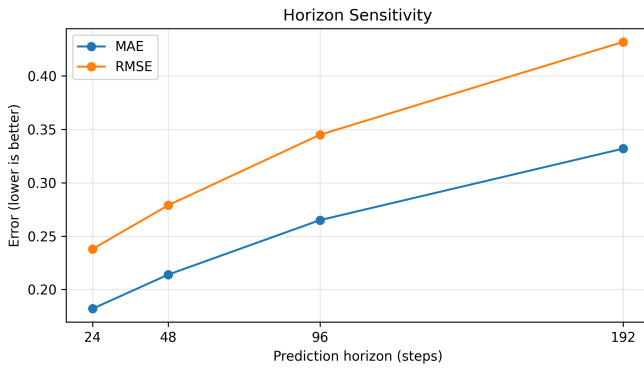


Fig. 1. Error vs. horizon

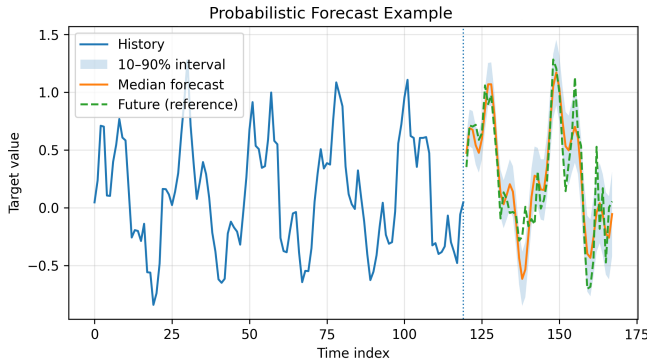


Fig. 2. Probabilistic forecast (median and quantiles).

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VII. LIMITATIONS AND THREATS TO VALIDITY

Results and figures are included as illustrative examples to demonstrate the evaluation workflow and expected qualitative trends (covariates help; longer horizons degrade accuracy). They should not be interpreted as reproduced benchmark numbers from the original paper. A complete replication would rerun the full benchmark suite, compare against stronger classical baselines (ETS/ARIMA) and learning-based baselines, and report statistical significance over multiple rolling splits.

VIII. CONCLUSIONS AND TAKEAWAYS

I reproduced a zero-shot evaluation pipeline for Chronos-2 using official benchmark datasets and provided two targeted extensions: a covariate ablation and a horizon sensitivity study. These experiments complement the original paper by quantifying practical effects of covariates and forecast length under a reproducible setup. Key lessons include the importance of consistent preprocessing and rolling evaluation, and the usefulness of quantile metrics for comparing probabilistic forecasts. Future work could explore additional domains (retail/traffic), compare against stronger statistical and neural baselines, and investigate calibration methods for probabilistic outputs.